

DATA Report: Bitcoin–Equity Linkages

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Key Data Processing Steps

- Cleaning the data and remove the duplicated sheets and Constructed the FX Converted Bitcoin leg:

$$r_{i,t}^k \approx \Delta \log(\text{BTC}_t^{USD}) + \Delta \log(S_{i,t}),$$

where $S_{i,t}$ is the LCU per USD. Which is consistent with the model defined in the paper.

- Computed local equity log returns for Brazil (IBOV), India (NIFTY), South Africa (JALSH), and the U.S. (SPXT) respectively.
- Created 60 day rolling correlations and ran baseline OLS regressions of r^s on r^k (daily data, 2020–2025).

Main Findings

- **Rolling correlations:** Pre 2020 near zero, Post 2020 persistently positive.
 - U.S.: mean ≈ 0.35 , peaks > 0.8 .
 - EMs: smaller but positive means (Brazil ≈ 0.04 , India ≈ 0.07 , South Africa ≈ 0.11).
- **OLS slopes (2020–2025, no controls):**

Table 1: Daily OLS since 2020: Local Equities on FX-BTC Leg (or SPXT on BTC, USD)

Market	(lhs)	$\hat{\beta}$	$t(\hat{\beta})$	R^2	Notes
Brazil (IBOV)	vs. r_{BRL}^k	0.048	5.34	0.014	positive, significant
India (NIFTY)	vs. r_{INR}^k	0.028	4.22	0.009	positive, significant
South Africa (JALSH)	vs. r_{ZAR}^k	0.053	7.45	0.026	positive, significant
U.S. (SPXT)	vs. BTC USD	0.135	18.84	0.145	larger magnitude

Market	$\hat{\beta}$	$t\text{-stat}$
Brazil (IBOV)	0.05	5.3
India (NIFTY)	0.03	4.2
South Africa (JALSH)	0.05	7.4
U.S. (SPXT)	0.14	18.8

In short, all significantly positive, with the U.S. slope largest.



Figure 1: 60-day rolling correlation between SPXT and BTC (USD), Jan 2020–Sep 2025. Notes: Daily log returns; 60-trading-day window. Positive and persistent co-movement emerges post-2020, consistent with the paper’s thesis.

Interpretation

All slopes are significantly positive post 2020, in line with the model’s directional prediction. However, raw daily $\hat{\beta}$ magnitudes are well below 1 in this un-controlled specification. This is not a contradiction of the theory: the model predicts $\beta \rightarrow 1$ after removing common shocks and when idiosyncratic wedge variance is small. Empirically, achieving a Near Unity Slope requires (i) controlling global risk and time effects, (ii) aligning frequencies, and (iii) using a Residual On Residual regression consistent with the parity decomposition.

Alignment with the Model and Implications

Consistency with the stylized facts. The rolling correlations and post-2020 OLS slopes corroborate a shift from near-zero (pre-2020) to persistently positive co-movement (post-2020), particularly strong in the U.S. EMs also turn positive on average with episodic spikes.

Bridging the theory to the regression. Let m_{t+1} denote the real LCU SDF and $w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}}$. From the parity,

$$\mathbb{E}_t[m_{t+1}((1 + R_{t+1}^{s,\text{real}}) - (1 + R_{t+1}^{k,\text{real}}))] = 0,$$

the regression of r^s on r^k identifies β as the slope of *idiosyncratic* components after purging common shocks. A practical implementation is to control X_t^{global} (VIX, dollar conditions, global rates, EMBI/GFCI) and include time fixed effects λ_t , or equivalently to regress out these factors from both r^s and r^k and then estimate $\hat{\varepsilon}^s$ on $\hat{\varepsilon}^k$.